

Build the AdHush Box

A little box that sits between your cable box and your TV, **watches for commercials, and clicks the sound off** until your show comes back. No soldering. About an afternoon. Around **\$120–160** in parts. If you can build a LEGO set from instructions and type commands carefully, you can build this.

NESPi 4 CASE EDITION — Pi 4 B (2 GB) in a Retroflag NESPi 4, 5 V relay

⚡ Safety rules — read these first

Everything in this project runs on **low-voltage USB power**. The biggest voltage you will ever touch is 5 volts — about the same as a phone charger cable.

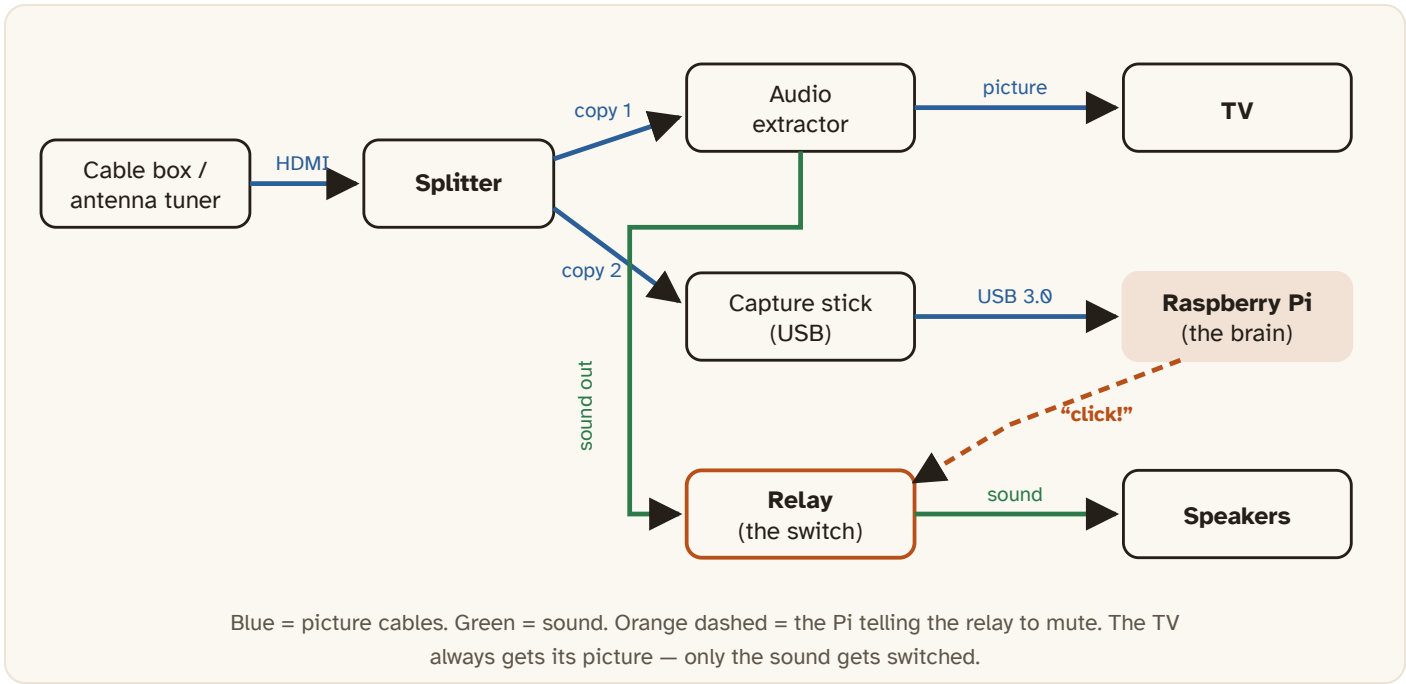
Never open, cut, or wire anything that plugs straight into a wall outlet. The only cable you cut in this whole build is a thin headphone-style audio cable, and it must be *unplugged from everything* while you cut it.

Where you see the **ADULT CHECK** tag, stop and have a grown-up look at your work *before* you plug in any power.

★ The big picture

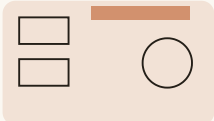
The TV signal gets **split into two copies**. One copy goes to your TV like normal. The other copy goes to the Raspberry Pi, which watches it. The **sound** takes a detour through a little switch called a **relay** — and the Pi clicks that switch whenever a commercial starts.

In this edition the Pi lives inside a **NESPi 4** case — the one that looks like an old Nintendo. That is nice (fan, heatsinks, buttons, and it looks great under a TV), but a closed case changes two things: **one thin wire has to leave the case**, and the relay gets its power from a **USB port** instead of a header pin. Everything else is the standard build.



— picture (HDMI / USB) — sound — control signal — 5 V power

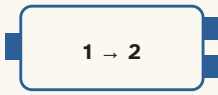
The parts



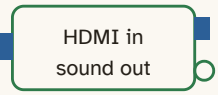
1 · Raspberry Pi 4



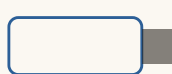
2 · microSD 32 GB



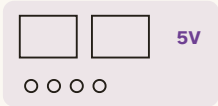
3 · HDMI splitter



4 · Audio extractor



5 · Capture stick



6 · Relay module
2-channel, 5 volt



7 · Jumper wires
kit + mini breadboard



8 · 3.5 mm audio
cables ×2 (one gets cut)



9 · Short HDMI
cables ×2



10 · Small screwdriver,
scissors, tape

Lay everything out before you start, like the first page of a LEGO manual. Checking the list now is much easier than finding a missing part halfway through.

Part	What it does	≈ Price
Raspberry Pi 4 B (2 GB) + official USB-C power supply	The brain — a tiny computer	\$45–60
Retroflag NESPi 4 case — comes with its fan, heatsinks, adapter board, and the SSD “cartridge”	The box the brain lives in	\$35–50
Micro-HDMI to HDMI cable (setup day only; some case bundles include one)	The Pi’s HDMI ports are the tiny kind, so the setup monitor needs this	\$8
microSD card, 32 GB	The brain’s memory	\$8

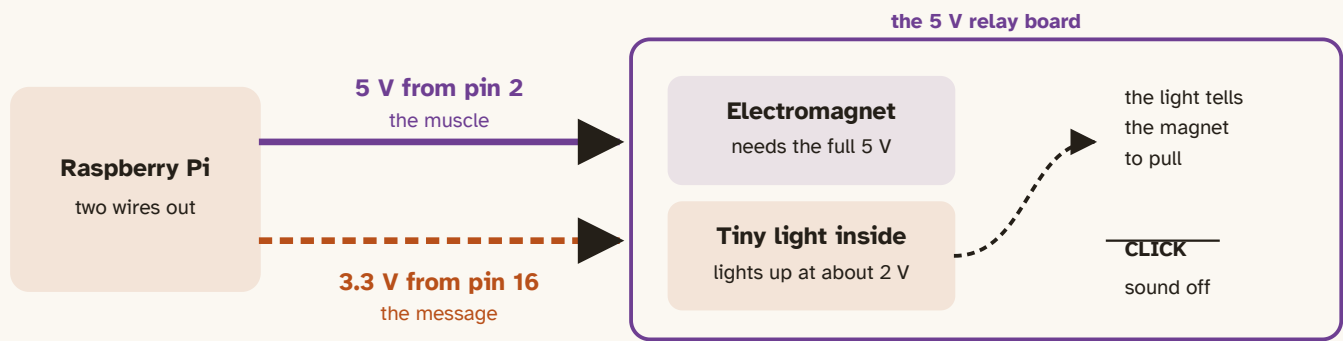
Part	What it does	≈ Price
Powered HDMI splitter, 1 in → 2 out	Makes a copy of the picture	\$15
HDMI audio extractor (HDMI in → HDMI out + headphone/RCA out)	Pulls the sound onto a normal audio cable	\$15–20
USB HDMI capture stick (must say “UVC”; USB 3.0 is best)	Lets the Pi see the copied picture	\$8–25
Relay module, 2-channel, 5 V, with screw terminals — opto-isolated is best. This is the kind hobby shops keep in stock.	The click-switch that cuts the sound	\$7
Jumper wire kit with a mini breadboard — you will use 2 female-to-female and 3 male-to-female wires	Connect the Pi to the relay, and send the one mute signal to <i>both</i> relay channels (Step 6 explains)	\$8–10
Two 3.5 mm audio cables (one gets cut in half) — or, to skip the cutting, two “3.5 mm plug to screw terminal” adapters	Carry the sound	\$6
2 short HDMI cables	Connect the boxes	\$10
USB-A male breakout with screw terminals (a USB plug on a little board with 4 screws)	Feeds 5 V from a case USB port to the relay — Step 6 explains why	\$3–5
Double-sided foam tape	Sticks the relay module to the back of the case	\$3
Small screwdriver, scissors, electrical tape	Tools you probably already have	—

Borrow for setup day: a keyboard, a mouse, a monitor or spare TV, and your Wi-Fi password. Speakers or a soundbar with an aux input sound best; your TV’s own “audio in” jack works too.

5V Why a 5 volt relay?

Older versions of this guide asked for a **3.3 volt** relay module, because 3.3 volts is what the Pi’s little signal pins put out. The trouble is that almost nobody sells 3.3 V modules in a store — walk into a hobby shop and every relay board on the shelf says **5 V**.

Good news: a 5 V board works, and this guide is written for it. Here is the trick — a relay board has **two separate jobs**, and they need different amounts of electricity.



The **5 V wire** does the heavy lifting. The Pi's 3.3 V signal wire only has to light a tiny lamp inside the board, and that lamp needs about 2 volts — so 3.3 volts is plenty. That is why a 5 V board works from a 3.3 V Pi.

What changed from the old 3.3V guide

	Old 3.3V guide	This 5V guide
Relay module	3.3 V logic (hard to buy)	5 V (on every shelf)
VCC wire	pin 1 — 3.3 V	pin 2 or pin 4 — 5 V
GND, signal, sound wiring	pin 6, pin 16, COM + NC	all unchanged
Settings	<code>active_high = true</code>	the same, unless the test comes out backwards

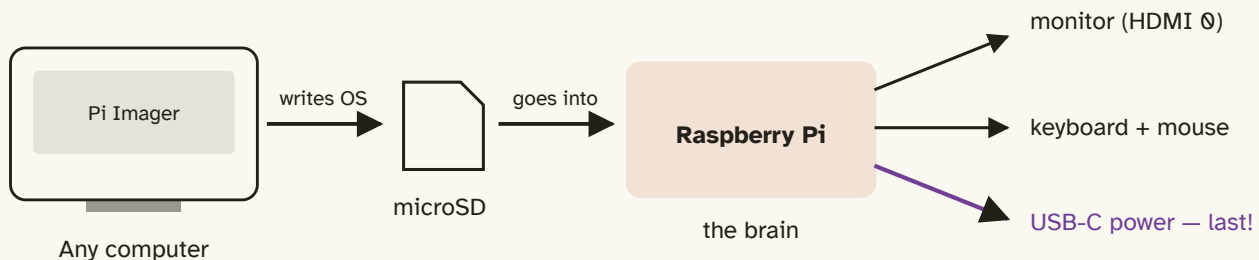
1

Wake up the Pi

ADULT CHECK

Do this step with the Pi bare on the table — before the case

Once the Pi is inside the NESPi 4, its USB ports are the two under the front flap, and this build uses both of them (capture stick + relay power). So do all of Step 1 — keyboard, mouse, monitor, the four typed lines — with the Pi sitting naked on the table. Then put it in the case in Step 2 and never open it again. The monitor plugs into the Pi's **micro-HDMI** port (the one nearer the USB-C power plug), so you need the micro-HDMI cable from the parts list.



Setup day only: the monitor, keyboard, and mouse all go back when you're done. Plug the power in *last* — a Pi turns on the moment it gets power.

1. On a computer, install **Raspberry Pi Imager** (free, from raspberrypi.com). Put in the microSD card and write **Raspberry Pi OS (64-bit)** to it — the plain, recommended one at the top of the list. In the imager's settings gear, set a username, a password, and your Wi-Fi name and password.
2. Move the card into the Pi. Connect the monitor, keyboard, and mouse. Power last. Wait for the desktop to appear.
3. Open the black **Terminal** window and type these four lines, pressing Enter after each one. Type them exactly — capital letters matter:

```
git clone https://github.com/socrtwo/adhush
cd adhush
sudo scripts/install-pi.sh
cp config/adhush-passthrough.example.toml config/adhush.toml
```

Three things about picking the OS

The wrong pick in that list wastes an evening, so read these before you click Write.

- **64-bit**, not 32-bit. AdHush is all number-crunching, and 64-bit is about 25% faster at it. More importantly, the maths library it needs has ready-made 64-bit downloads — on 32-bit the Pi may have to *build* that library itself, which takes about an hour and can run out of memory on a 2 GB board.
- **Not** the one labelled **Legacy**. That one is older and comes with Python 3.9. AdHush needs 3.11 or newer, so the install would stop with an error on your very first command.
- **Desktop**, not **Lite**. Lite has no desktop at all, so there would be no Terminal window to open in step 3.

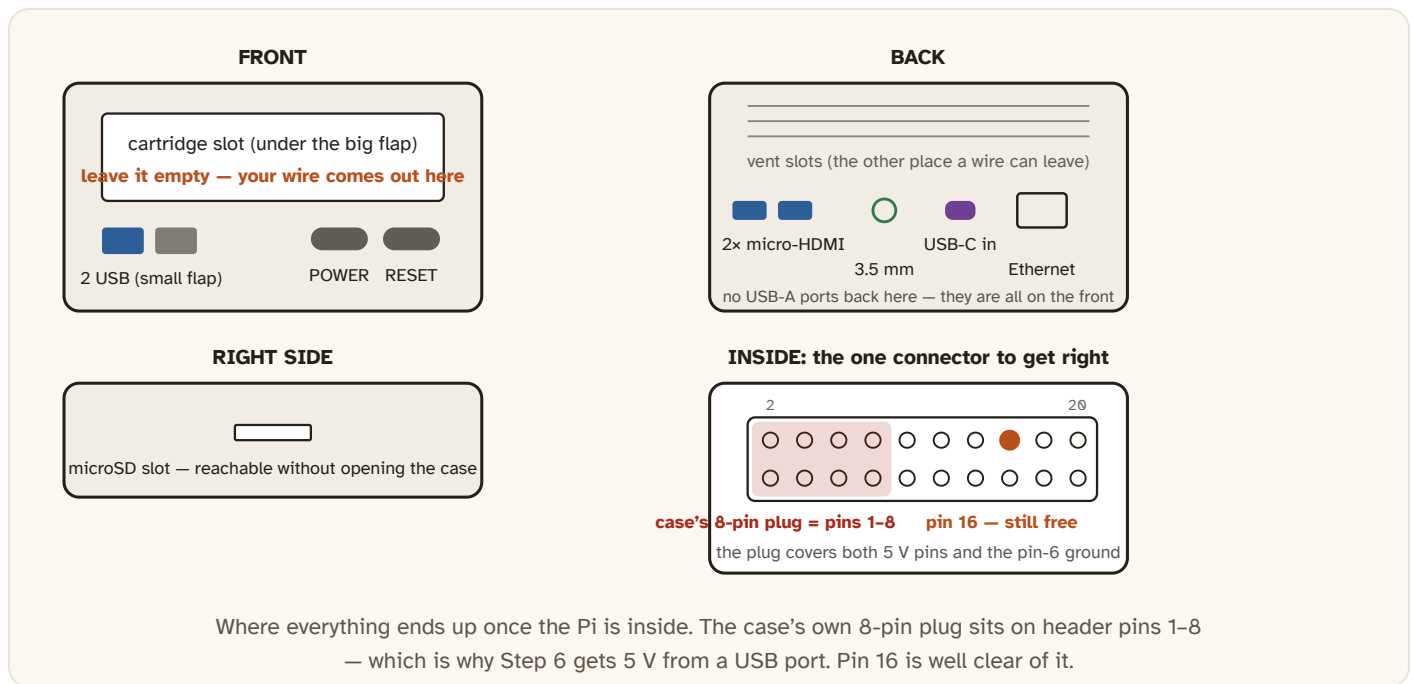
That last line makes the box's settings file. Leave it alone for now — the defaults match this build. You'll come back to it in Step 7 only if the relay turns out to click backwards.

2

Build the Pi into the NESPi 4

ADULT CHECK

The case comes with its own little manual and screws; follow that for where each screw goes. These are the parts of the build that matter for AdHush, and the three settings you must get right.

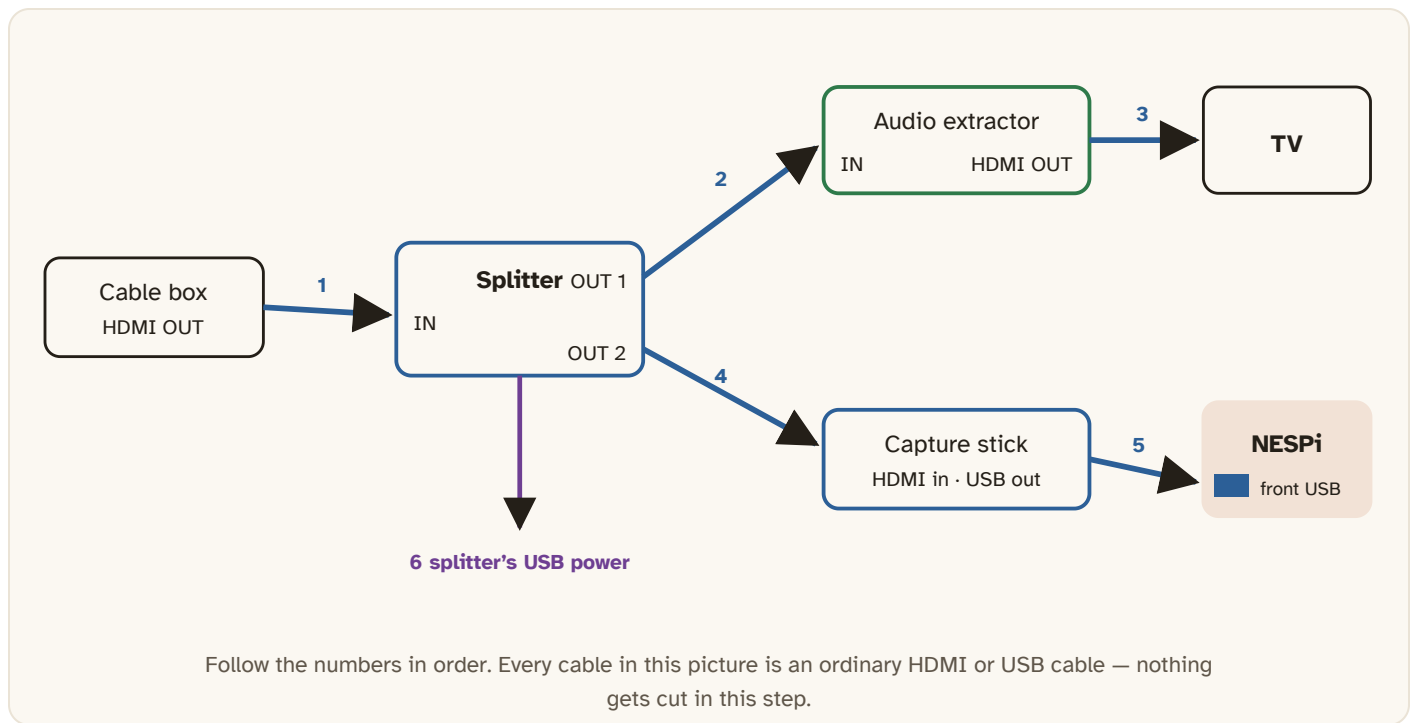


1. Stick the **heatsinks** from the case box onto the Pi's two biggest chips. AdHush works the Pi hard all evening, so these matter.
2. Follow the case manual to screw the Pi in and plug its USB and Ethernet ports into the case's **adapter board**. Plug the **fan** into the fan header on the case board.
3. Push the case's **8-pin plug** onto header pins **1-8**: the two rows of four at the corner nearest the microSD slot. The manual shows which way round. **Count the pins**: it must not sit one column over.
4. Set the case board's **SAFE SHUTDOWN** switch to **OFF**. The ON position needs Retroflag's helper script, which no longer works on current Raspberry Pi OS. OFF turns the POWER button into a plain power switch — fine for a box that stays on.
5. Leave the **cartridge empty** — no SSD. AdHush needs very little storage, and the empty slot is the neatest way for your one wire to leave the case.
6. Close the case, but don't screw the lid down until Step 6 is done and tested.

The NESPi 4 changed a little over its production run: later ones have a blue **USB 3.0** port under the front flap, earlier ones have two black USB 2.0. Either works for this build — the capture stick runs at 720p30, which fits USB 2.0 fine.

3 Build the picture path

No tools needed — this step is all plugging in cables. You are making the copy of the TV signal that the Pi will watch.



1. Unplug the HDMI cable that runs from your **cable box to your TV**, and plug that end into the **splitter's IN** instead.
2. New cable: splitter **OUT 1** → audio extractor **IN**.
3. New cable: extractor **HDMI OUT** → your **TV**. (Picture is back!)
4. New cable: splitter **OUT 2** → the **capture stick**.
5. Plug the capture stick into a **USB port under the case's front flap** — the blue one if yours has one; a black one works too.
6. Connect the splitter's little USB power cord.

Test it: turn on the TV. You should see your channels exactly like before. Black screen? Try the splitter's other output, or swap in a different HDMI cable.

Then tell the Pi which gadget is the sound

The Pi can hear from more than one place — its own headphone jack, the capture stick, anything else that is USB. It needs to know which one is the stick. In the terminal, inside the `adhush` folder, type:

```
adhush doctor
```

Near the bottom it prints a list of **sound cards**. It looks like this:

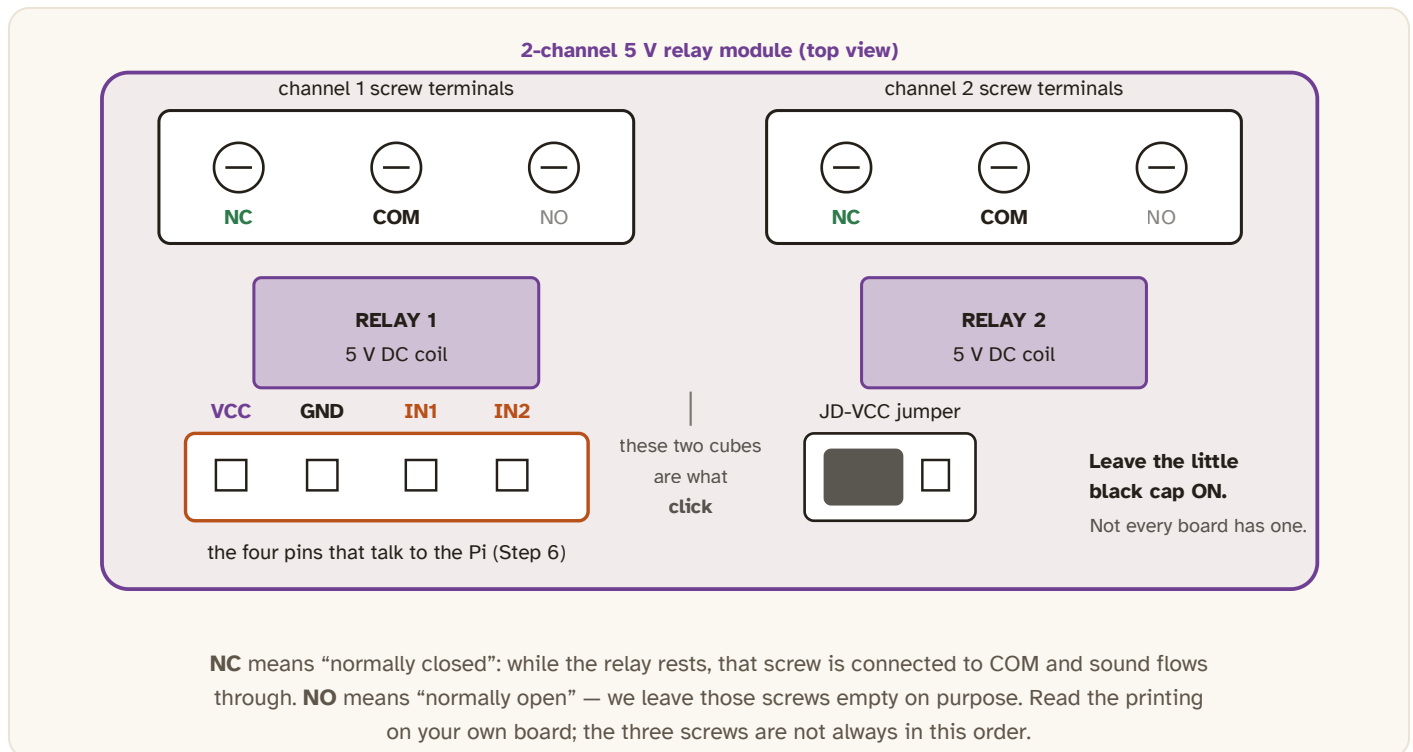
```
sound cards – put the capture stick's card in capture.audio_device:
  card 0: Headphones      bcm2835_Headpho - bcm2835 Headphones
           use alsa:hw:0,0 (or alsa:hw:CARD=Headphones,DEV=0, which survives reboots)
  card 1: MS2109          USB-Audio - MS2109 <- USB, probably the stick
           use alsa:hw:1,0 (or alsa:hw:CARD=MS2109,DEV=0, which survives reboots)
video devices – capture.device: /dev/video0 /dev/video1
```

1. Find the card marked **USB, probably the stick** and read its number. On most Pis it is **card 1** — and that is what the settings file already says, so you are done.
2. If your number is different: open the settings with `nano config/adhush.toml`, find the line that starts `audio_device =`, and change the `1` in `"alsa:hw:1,0"` to your number. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
3. Run `adhush doctor` again. The line `audio device alsa:hw:... present` should now say `ok`.

If the box ever goes deaf after a reboot, USB gadgets sometimes swap numbers. Paste the name form instead — the one like `alsa:hw:CARD=MS2109,DEV=0` — and it stays put.

4 Meet your relay board

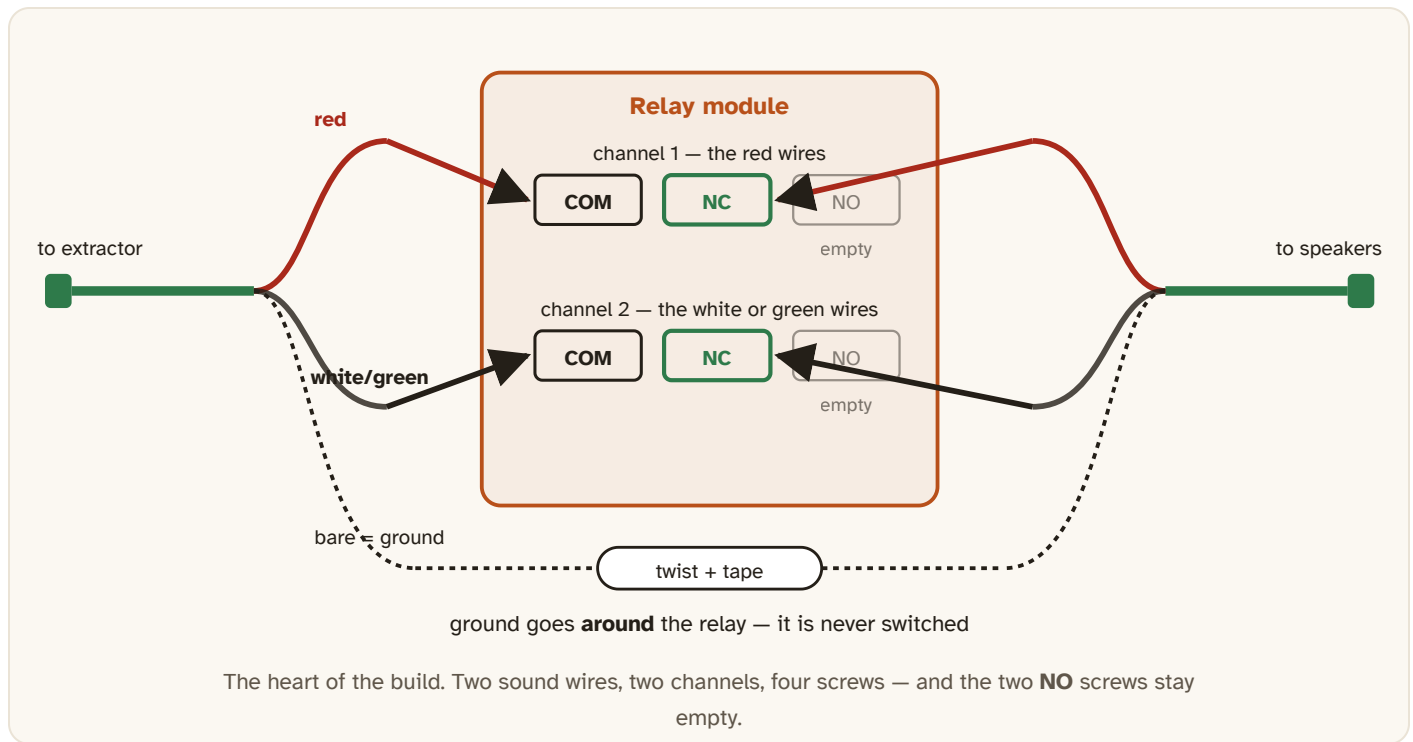
Before any cutting, get to know the board. A 2-channel 5 V relay module has **two sides**: a *low-voltage side* where four little pins talk to the Pi, and a *switch side* with fat screw terminals where your sound wires go. The two sides never touch electrically — that is exactly why this is safe.



The one rule that makes this safe to leave running

Sound goes through **COM + NC**, never NO. That way, if the Pi crashes, loses power, or you unplug the whole thing, the relay falls back to resting and your sound comes **back on**. A broken AdHush box should be a box that does nothing — not a box that leaves your TV silent.

Now the scissors. Sound will travel: extractor → **through the relay** → speakers. Make sure the cable you are about to cut is unplugged from everything first.



1. Cut one audio cable in the middle. Strip about 2 cm of the outer cover off each cut end. Inside are three small wires: one **red**, one **white or green**, and a bare copper one. The bare one is **ground**. (Some cables use a copper mesh instead; gather it and twist it into a strand.)
2. Look closely at the strands. If they are shiny and look painted, they are — scrape the last 1 cm with sandpaper or the back of a knife until dull copper shows. Painted strands will not make contact in a screw terminal.
3. Twist the two **ground** wires together and wrap the joint in tape. Ground is never switched.
4. Screw the **red** wires into channel 1's **COM** and **NC**. It does not matter which red goes to which of those two.
5. Screw the **white/green** wires into channel 2's **COM** and **NC**. Leave every **NO** screw empty.
6. Give each wire a gentle tug. If one pulls out, the screw isn't tight enough.
7. Plug one half of the cut cable into the **extractor's audio out** and the other half into your **speakers**.

No-scissors option

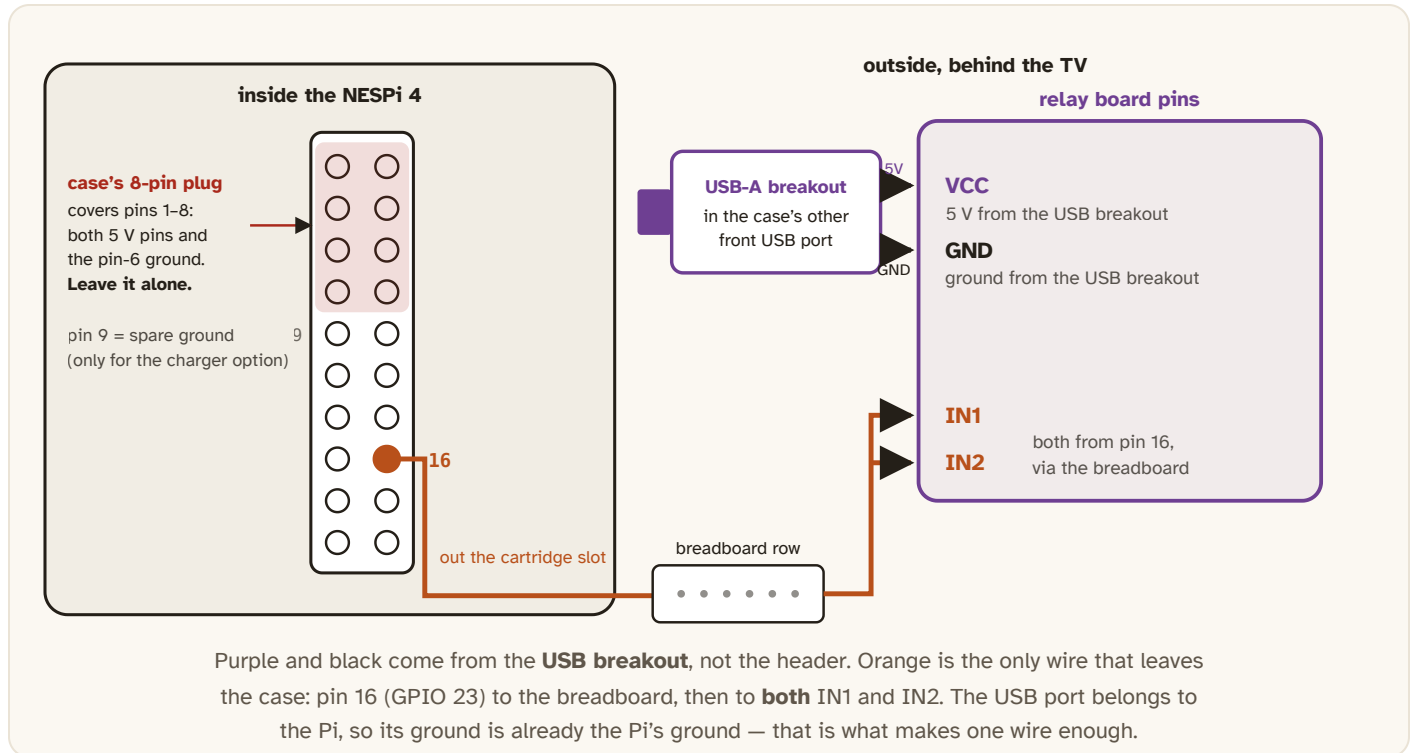
Two "**3.5 mm plug to screw terminal**" adapters (about \$6 a pair) do the same job as the cut cable, with nothing to cut or scrape. Plug one adapter into the extractor and one into the speakers. Each has three screws: **L**, **R**, and **G** (ground). Run short wires L→relay channel 1→L, R→channel 2→R, and join the two G screws to each other with one wire. That is the whole sound path.

6

Connect the Pi to the relay

ADULT CHECK

This is the step the case changes. In the standard build the relay gets its 5 V from header **pin 2** and its ground from **pin 6**. In the NESPi 4 those pins are **under the case's own 8-pin plug**. So the relay gets 5 V and ground from a **USB port** instead, and only *one* wire — the mute signal from **pin 16** — has to come out of the case.



From	What it is	Goes to
USB breakout "5V" screw	5 V from a case USB port (may be marked VCC or V+)	VCC
USB breakout "GND" screw	Ground — shared with the Pi through the port	GND
Pi pin 16	GPIO 23 — the mute signal, 3.3 V — the one wire that leaves the case	IN1 and IN2
Pi pins 1–8	under the case's plug — both 5 V pins and the pin-6 ground live here	nothing — leave alone

1. Stick the relay module to the **back of the case** with foam tape, near the cartridge end. Stick the mini breadboard next to it.
2. Screw two short wires into the USB breakout's **5V** and **GND** screws. Ignore its D+ and D– screws. Run 5V to the relay's **VCC** and GND to the relay's **GND**. Do *not* plug the breakout in yet.
3. Open the case lid. Push a **male-to-female** jumper's female end onto **pin 16**: on the even-numbered row, the **eighth** pin along, counting from the case's plug end. Count twice.
4. Route that wire out through the **empty cartridge slot** (or a rear vent slot) and push its male end into any breadboard row. Close the lid; leave the screws for after the test.
5. Two more male-to-female jumpers from that **same breadboard row** to the relay's **IN1** and **IN2**.

6. Now plug the USB breakout into the case's **other front USB port**. The relay board's little power light should come on.

If you'd rather power the relay from a phone charger

That works too, but then the relay's ground is no longer automatically the Pi's ground, and the signal needs a shared ground to mean anything. Add a second jumper from Pi **pin 9** (a free ground, just past the case's plug on the odd row) to the relay's **GND**. Two wires out of the case instead of one.

ADULT CHECK before power-on

Read this list out loud with a grown-up, pointing at each thing:

- The sound wires are on **COM** and **NC**. Both **NO** screws are empty. The two ground wires are twisted together and taped.
- The case's 8-pin plug is on pins **1–8** and nothing else touches it.
- The only jumper on the Pi's header is on **pin 16**.
- The USB breakout's **5V** goes to relay **VCC**, **GND** to **GND**, and its D+ / D– screws are empty.
- No bare metal is touching any other bare metal, inside the case or out.
- The little black **JD-VCC cap**, if your board has one, is still on.

Then power up: the case's USB-C, then its POWER button. You should hear the TV through your speakers. Turn the TV's own volume all the way down — the relay path is the sound now.

7 Wake it up

In the Pi's terminal, inside the `adhush` folder, run these one at a time. Start with the relay test — it is the fastest way to know your wiring is right.

```
adhush probe --active    # sound should cut out for 2 seconds, then come back
adhush calibrate         # run this while a SHOW is on, not an ad
adhush run               # start watching!
```

During `adhush probe --active` you should hear one *click*, two seconds of silence, then another *click* and the sound returns. That is the whole build, proven.

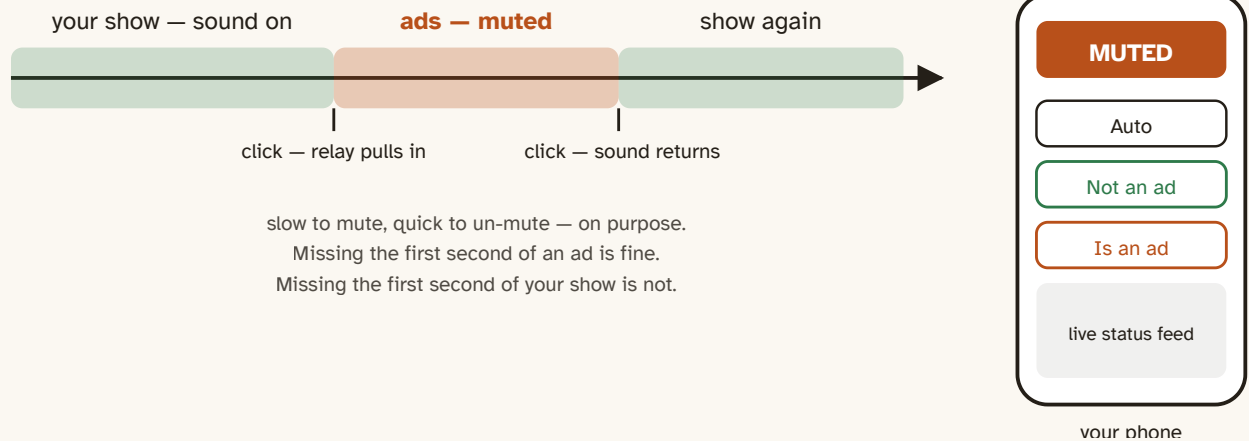
If the relay works backwards – the one 5V gotcha

Some 5 V boards mute when the signal goes *high*. Others mute when it goes *low*. You cannot tell from the outside, so you test. If your sound is **off at rest** and comes **on** during the two-second test, your board is the other kind. One-line fix, not a wiring mistake.

Open the settings file: `nano config/adhush.toml`, find the `[control.relay_hdmi]` part, and change `true` to `false`:

```
[control.relay_hdmi]
gpio = 23
active_high = false
```

Save with **Ctrl+O**, then Enter, then **Ctrl+X** to leave. Run `adhush probe --active` again — now it should be the right way round.



What success sounds like — and your phone remote. Open `platforms/web/index.html` on any device on your Wi-Fi to watch PROGRAM / MUTED live and press the two teaching buttons.

- **“Not an ad”** — press it if the box mutes your actual show. Sound comes back instantly *and* it forgets the mistake.
- **“Is an ad”** — press it during a commercial it missed, and it memorizes that ad for next time.
- The first evening it is mostly learning, and it gets noticeably sharper on ads it has heard before. To make the box start by itself whenever it has power, run `sudo systemctl enable --now adhush`.
- **Turning it off:** with SAFE SHUTDOWN set to OFF, the case’s POWER button just cuts the power. That is safe for the relay (sound comes back) but not kind to the microSD card if you do it often. Run `sudo shutdown -h now` from a remote shell first, wait for the green light to stop blinking, then press the button.

If something's wrong

Problem	Try this
No picture on the TV	Reseat the HDMI cables; check the splitter's power; swap OUT 1 and OUT 2.
No sound at all	Are both wires on COM + NC (not NO)? Are the grounds twisted together? Is each cable half plugged into the extractor and the speakers?
Never hears anything, or <code>doctor</code> says the audio device FAILs	Wrong card number. Re-read the sound-card list in <code>adhusht doctor</code> and fix <code>audio_device</code> (end of Step 3).
Relay LED is off, never clicks	The USB breakout is in a port with no power, or 5V/GND are swapped on its screws. Try the other front port; check the markings.
Case fan never spins	Fan plug on the case board's fan header, not the Pi. Red wire to +.
POWER button does nothing, or the Pi dies the instant you press it	That is the SAFE SHUTDOWN switch. OFF = plain power switch (what this build uses). ON needs Retroflag's script, which is broken on current Raspberry Pi OS.
Sound only on one side	Only one channel is wired, or one screw is loose. Channel 1 carries red, channel 2 carries white/green.
Muted at rest, sound during the test	Your board is the other kind. Set <code>active_high = false</code> — see the <i>relay works backwards</i> box in Step 7.
Relay never clicks at all	Run <code>sudo systemctl start pigpiod</code> , then <code>adhusht probe --active</code> again. Check VCC is on pin 2 , not pin 1 — a 5 V board will not pull in on 3.3 V.
Relay buzzes or chatters	Not enough current. Use the official Pi power supply and keep the JD-VCC cap on. If it still buzzes, that board wants a 3.3 V-logic module instead — not your fault.
It mutes my show!	Press Not an ad on the phone page — it learns from that.
<code>adhusht run</code> shows an error	Read the message, then run <code>adhusht doctor</code> — it lists exactly what is missing.

One honest note. A few channels and streaming boxes copy-protect their HDMI signal, so the capture stick sees nothing. AdHush never tries to break that protection. For those sources the box still works in listen-only mode, using just the sound to spot ads.

AdHush · NESPi 4 edition — open source under the MIT license · github.com/socrtwo/adhusht

Plain-text version: [docs/build-guide-beginner.md](#) · wiring detail and the reasoning behind the fail-with-sound design: [docs/hardware-passthrough-box.md](#) · building with an Arduino, ESP32, or Pico instead: [docs/build-guide-microcontrollers.md](#).

AdHush mutes your own speakers in your own home. It does not modify, copy, record, or redistribute anything you watch, and it never tries to defeat content protection.